Power BI Data Model

Step-by-Step Assignment Walkthrough

Building a Star Schema from a Flat Dataset in Power Query & Model View

📌 Tutor's Core Rules for Data Modeling

- **Rule 1:** Separate all reporting fields into Categories and Values.
- **Rule 2:** All Value (numeric) fields stay in the **Fact table**.
- **Rule 3:** Look for hierarchies in categorical fields. Categorical fields WITHOUT a hierarchy stay in the Fact table. Categories forming a hierarchy become a **Dimension table**.
- **Rule 4:** Identify the smallest unit in each dimension as its **Primary Key**.
- **Rule 5:** Replicate all Primary Keys in the Fact table to build relationships.
- **CRITICAL:** A Dimension table MUST contain 2 or more columns forming a real hierarchy — a single flat column is NOT allowed to be its own dimension.

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Course: Data Analytics Course | **Group:** King's House
Scope: Power Query Transformation to Star Schema Relationship Modeling (Excludes Visuals & Measures).

PART 1 — GET THE DATA INTO POWER BI

1. Open Power BI Desktop on your computer.
2. In the **Home** tab on the top ribbon, click **Get Data** → select **Excel Workbook**.
3. Browse your file folders, select `CLEANED_DATA.xlsx`, and click **Open**.
4. In the **Navigator** window that pops up, tick the checkbox next to **"CLEAN"**. Then click **Transform Data** (*DO NOT click "Load" — clicking Transform Data opens Power Query Editor where we build the model structure*).

PART 2 — SET UP THE BASE QUERY

5. You are now inside **Power Query Editor**. On the left-hand panel (Queries Pane), you will see a query named **"CLEAN"**. This represents your raw, untransformed data source.
6. Verify that Power BI auto-detected correct data types (e.g., calendar icons next to dates, '123' icons next to numeric values) under the **"Changed Type"** step in Applied Steps. No action needed unless a column type is wrong (e.g., a date saved as text).

PART 3 — BUILD THE CUSTOMER DIMENSION (DCUSTOMER)

7. In the left Queries panel, right-click on **"CLEAN"** and click **Reference**. (*This creates a new query linked to CLEAN without altering the original raw data query*).
8. Right-click this newly created query → click **Rename** → type `DCUSTOMER` and press Enter.
9. Click into the **DCUSTOMER** table view. Hold down the `Ctrl` key and click the column headers for **"Customer_ID"** and **"Customer_Segment"** to select both.
10. Right-click either selected column header → click **Remove Other Columns**.
11. Right-click the **"Customer_ID"** column header → click **Remove Duplicates**.

> ✅ **Result:** DCUSTOMER now contains exactly 2 columns (Customer_ID, Customer_Segment) with 1 unique row per customer.

PART 4 — BUILD THE PRODUCT DIMENSION (DPRODUCT)

12. Right-click **"CLEAN"** in the left Queries panel → click **Reference**.
13. Right-click the new query → **Rename** it to `DPRODUCT`.
14. Hold down `Ctrl` and click the column headers for **"Category"** and **"Product"** to select both columns.
15. Right-click either selected header → select **Remove Other Columns**.
16. With both columns still highlighted, right-click either header → select **Remove Duplicates**.

> ✅ **Result:** DPRODUCT now contains exactly 2 columns (Category, Product) with 1 unique row per product.

PART 5 — BUILD THE CALENDAR DIMENSION

17. In the left Queries panel, right-click in any empty space → **New Query** → **Blank Query** (or click Home tab → New Source → Blank Query).
18. Right-click the new query → **Rename** → type CALENDAR.
19. With **CALENDAR** selected, go to the **Home** tab on the top menu → click **Advanced Editor**.
20. Delete all existing code in the dialog box and paste this M-code snippet exactly:

```
let Source = List.Dates(#date(2025,1,1), 365, #duration(1,0,0,0)), ToTable = Table.FromLis
```

21. Click **Done**. You will now see a continuous date list of 365 rows for 2025.

    💡 ***Note to classmates:*** *Double-check the Order_Date range in your CLEAN query. If your date range differs from Jan 1–Dec 31, 2025, edit the start date and day count in the code above.*

22. Go to **Add Column** tab → click **Date** → hover over **Year** → click **Year**.
23. Click the original **Date** column again → **Date** → hover **Quarter** → click **Quarter of Year**.
24. Click **Date** again → **Date** → hover **Month** → click **Name of Month**.
25. Click **Date** again → **Date** → hover **Month** → click **Month** (*this adds the month NUMBER 1–12, required so month names sort chronologically instead of alphabetically in reports*).

> ✅ **Result:** CALENDAR has 5 columns (Date, Year, Quarter, Month Name, Month) and 365 continuous rows.

PART 6 — BUILD THE FACT TABLE (ORDERS)

26. Right-click **"CLEAN"** in the Queries panel → select **Reference**.
27. Right-click the new query → **Rename** it to ORDERS.
28. Right-click the **"Category"** column header → click **Remove** (*removed because it now resides in DPRODUCT*).
29. Right-click the **"Customer_Segment"** column header → click **Remove** (*removed because it now resides in DCUSTOMER*).
30. Leave every other column intact — including Foreign Keys (Customer_ID, Product, Order_Date), Region, and all numeric metric columns.

> ✅ **Result:** ORDERS now contains exactly 21 columns representing core facts and key fields.

PART 7 — STOP THE RAW "CLEAN" QUERY FROM LOADING

31. Right-click **"CLEAN"** in the left Queries panel → click **Properties** (or untick "Enable Load" directly in the context menu).
32. Untick **"Enable Load"** → click **OK**. (*CLEAN's name will turn italicized, meaning it powers reference tables behind the scenes but won't load into the data model*).

PART 8 — LOAD EVERYTHING INTO THE MODEL

33 Click the **Close & Apply** button at the top-left corner of Power Query Editor. Power BI will now process transformations and load 4 clean tables into the model: DCUSTOMER, DPRODUCT, CALENDAR, and ORDERS.

PART 9 — BUILD THE DATA MODEL (RELATIONSHIPS)

34 On the far-left dark vertical toolbar, click the bottom icon resembling a diagram with connected boxes to enter **Model view**.

35 Click and drag each table diagram box by its title bar so all 4 tables are clearly visible and spread out in a Star Schema layout (Fact table in the center, Dimensions surrounding it).

36 **Connect DCUSTOMER to ORDERS:** Click and hold `Customer_ID` inside the **DCUSTOMER** box, drag your mouse over to `Customer_ID` inside the **ORDERS** box, and release.

37 **Connect DPRODUCT to ORDERS:** Click and hold `Product` inside the **DPRODUCT** box, drag onto `Product` inside the **ORDERS** box, and release.

38 **Connect CALENDAR to ORDERS:** Click and hold `Date` inside the **CALENDAR** box, drag onto `Order_Date` inside the **ORDERS** box, and release.

39 If a relationship settings dialog pops up, confirm that **Cardinality** is set to **One to many (1:*)** and **"Make this relationship active"** is ticked. Click **Save / OK**.

> ✅ **Result:** 3 relationship lines link DCUSTOMER, DPRODUCT, and CALENDAR to ORDERS — completing your Star Schema!

PART 10 — SAVE YOUR WORK

40 Press `Ctrl` + S, choose your save location, and name your file clearly (e.g., `KingsHouse_PowerBI_Assignment.pbix`).

⚠️ Avoid These Common Mistakes

- **Creating a 1-column Customer dimension:** Making a Customer table with ONLY `Customer_ID` violates our tutor's rule. A Dimension table must contain at least 2 columns forming a hierarchy (e.g., Customer_ID and Customer_Segment).
- **Deduplicating Order_Date for Calendar:** Generating a calendar table by removing duplicates from `Order_Date` misses days with zero sales (e.g. 362/365 days). You MUST use a complete 365-day continuous list.
- **Leaving CLEAN loaded:** Forgetting to disable "Enable Load" on the raw `CLEAN` query leaves a large, redundant disconnected table in your Data Model.
- **Connecting mismatched fields:** Dragging `Order_Date` instead of `Date` from the CALENDAR table into ORDERS breaks date intelligence.

✅ Final Submission Checklist

- [✓] **4 Tables Loaded (DCUSTOMER, DPRODUCT, CALENDAR, ORDERS)**
- [✓] **Raw CLEAN Query Italicized (Enable Load OFF)**
- [✓] **3 Active One-to-Many Relationships Connected**
- [✓] **File Saved as .pbix**